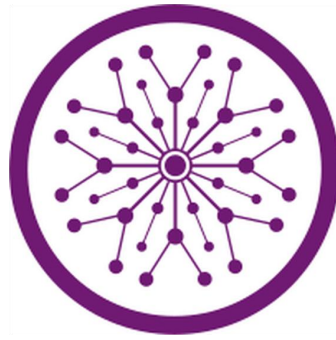


An Agentic AI Framework for Intelligent Patient Monitoring and Clinical Decision Support



SUPERIOR UNIVERSITY

A Thesis submitted in partial fulfillment
of the requirements for the degree of

MASTER OF SCIENCE
IN
ARTIFICIAL INTELLIGENCE

Submitted By

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No part of this thesis has been submitted anywhere else to any other degree. This thesis is submitted to the “Faculty Of Computer Science And Information Technology”, The Superior University, Lahore, in partial fulfillment of the requirements for the degree.

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DEDICATION

To my family.

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LIST OF ABBREVIATIONS AND ACRONYMS

AI	Artificial Intelligence
CDSS	Clinical Decision Support System
CoT	Chain-of-Thought
EHR	Electronic Health Record
HITL	Human-in-the-Loop
ICU	Intensive Care Unit
LLM	Large Language Model
MIMIC	Medical Information Mart for Intensive Care
RAG	Retrieval-Augmented Generation
ReAct	Reasoning and Acting
XAI	Explainable Artificial Intelligence

ABSTRACT

Clinical teams in intensive care generate vast, heterogeneous data, including vital signs, laboratory results, medications, and free-text notes, yet turning this stream into timely, trustworthy decisions remains difficult. Large language models (LLMs) answer medical questions well but behave as passive responders: they lack persistent patient memory, autonomous planning, and verifiable grounding, and they are typically evaluated on static examination-style questions rather than on the longitudinal record of a real patient. This thesis designs an Agentic AI framework for intelligent patient monitoring and clinical decision support that addresses these limitations. The framework organizes specialized LLM agents for monitoring, diagnosis, risk prediction, treatment recommendation, explanation, and a verification gate under a coordinator that routes tasks by patient condition and resolves conflicting recommendations. It couples the ReAct reasoning paradigm with retrieval-augmented generation (RAG) that grounds every recommendation in both the patient’s own electronic health record timeline and external clinical guidelines, and it preserves longitudinal context through a layered memory model. A cross-cutting trustworthy-AI layer provides explainability, audit logging, bias monitoring, confidence calibration, and human-in-the-loop validation so that clinicians retain final authority. The publicly available MIMIC-IV critical-care database is used as the data source, and the study specifies a bounded prototype together with an evaluation protocol that compares the framework against single-LLM, guideline-only RAG, and multi-agent baselines using measures of diagnostic accuracy, factual grounding, risk-prediction discrimination, safety, explanation faithfulness, calibration, and runtime cost. The expected contribution is a coherent, verifiable architecture for longitudinal clinical decision support, an explicit account of how patient-timeline retrieval and a dedicated verification agent affect hallucination and safety, and a reproducible design that future work can implement and validate. The thesis is positioned as a design and methodology contribution that lays the foundation for empirical clinical evaluation.

Keywords: Agentic AI; large language models; multi-agent systems; retrieval-augmented generation; clinical decision support; patient monitoring; MIMIC-IV; trustworthy AI.

CHAPTER ONE

INTRODUCTION

This chapter introduces the research and sets out its motivation, aims, and structure. It opens with the background that frames clinical data overload and the shift from passive Large Language Models to Agentic AI, then states the problem the thesis addresses and the specific gap it targets. The chapter next presents the research objectives and the five research questions that guide the investigation, defines the scope and boundaries of the work, and explains its significance for both research and clinical practice. It then gives an overview of the design-oriented research methodology, describes how the remaining chapters are organized, and closes with a summary of the argument developed here. Together these sections establish why an integrated, supervised Agentic AI framework for intelligent patient monitoring is needed and how the rest of the thesis develops it.

1.1 Background

Modern hospitals produce clinical data faster than clinicians can read it. A single intensive care admission generates streams of vital signs, laboratory panels, medication orders, imaging reports, and free-text notes, and each of these has to be interpreted under time pressure and often at odds with an already heavy caseload. The bottleneck is no longer the availability of data but the cognitive work of turning it into a decision. This tension has made healthcare one of the most studied application areas for artificial intelligence, where the promise is not to remove the clinician but to reduce the load of synthesis that precedes every judgment [1].

Large Language Models (LLMs) sharpened that promise considerably. They read clinical narratives, answer medical questions, and summarize records with a fluency that earlier natural-language systems never reached, and specialized models such as Med-PaLM have shown that this competence extends to expert-level medical question answering [2][1]. Yet fluency is not the same as competence in a clinical workflow. A conventional LLM assistant waits for a prompt, answers once, and forgets the exchange. It holds no persistent record of the patient, plans nothing beyond the current turn, and cannot consult an external guideline unless a human hands it one. For episodic question

answering these limits are tolerable; for continuous patient monitoring, where context accumulates over days and a missed trend can matter, they are disqualifying.

Agentic AI reframes the model as one component inside a larger loop rather than the whole system. An agent perceives its environment, reasons about a goal, plans a sequence of steps, calls external tools, writes to and reads from memory, and revises its approach when feedback contradicts its expectations [3][4]. The reasoning-and-acting pattern formalized in ReAct, the multi-agent coordination demonstrated by AutoGen and CAMEL, and the emergent social behavior of Generative Agents each supplied a piece of this machinery [5][6][7][8]. Sapkota and colleagues argue that the shift from a single tool-using model to a coordinated society of specialized agents is a genuine architectural break rather than a rebranding, and that distinction matters for a domain as compartmentalized as medicine [9].

Healthcare-oriented agent systems have started to appear, and they are instructive precisely because they remain partial. MedAgents stages a role-played consultation among specialist agents to improve medical reasoning [10]; MedRAG couples retrieval with structured medical knowledge to raise diagnostic specificity [11]; Agent Hospital simulates an entire care setting populated by autonomous agents [12]; and Clinical Camel adapts an open model for medical dialogue [13]. Each advances one capability well. None assembles reasoning, persistent patient memory, evidence retrieval, multi-agent collaboration, and physician oversight into a single architecture aimed at longitudinal monitoring, and that omission is the space this thesis occupies.

This research proposes an Agentic AI framework for intelligent patient monitoring and clinical decision support. The proposed framework organizes memory management, Retrieval-Augmented Generation (RAG), multi-agent collaboration, reasoning, planning, and trustworthy-AI safeguards into a layered design, and it uses the publicly available MIMIC-IV critical-care database as its data source [14][15]. The design goal is deliberately modest about autonomy and firm about accountability: the framework generates explainable recommendations, but a clinician reviews and authorizes every one of them through a human-in-the-loop step.

1.2 Problem Statement

Healthcare organizations accumulate structured and unstructured records at a scale that outpaces manual interpretation. Electronic Health Records (EHRs), laboratory reports,

medication histories, physiological measurements, and clinical notes each carry part of the picture, and a clinician must reconcile them quickly to reach a defensible decision about diagnosis, treatment, and ongoing management. Volume and heterogeneity together make that reconciliation slow and error-prone, and the cost of the delay falls hardest in critical care.

LLM-based tools address only the surface of this problem. They read and generate clinical language well, but most deployed systems behave as passive responders: they answer the prompt in front of them and retain nothing afterward [1]. They rarely plan across steps, seldom maintain a durable model of the patient, and do not, on their own, coordinate with other specialized components. The consequence is a landscape of narrow tools, each solving one prediction or extraction task, with little that supports a clinical workflow end to end [16].

Existing clinical decision support inherits these weaknesses and adds its own. Systems that achieve strong predictive accuracy often cannot explain a recommendation, cannot retrieve and cite current medical evidence, and cannot preserve long-term patient context across encounters. Where retrieval is added, as in RAG-based medical systems, it is typically applied to isolated question answering rather than to continuous monitoring under supervision [11][17]. What is missing is not another accurate classifier but an integrated architecture.

This thesis therefore targets a specific gap: the absence of a unified Agentic AI framework that combines memory management, reasoning, planning, Retrieval-Augmented Generation, multi-agent collaboration, and trustworthy-AI principles for intelligent patient monitoring, and that is designed and evaluated against real-world critical-care data such as MIMIC-IV [14].

1.3 Research Objectives

The primary objective is to design an Agentic AI framework for intelligent patient monitoring and clinical decision support, developed and assessed using the MIMIC-IV dataset [14].

The specific objectives follow from it. The first is to survey recent work on Agentic AI, LLMs, and intelligent healthcare systems, and to read that literature critically rather than descriptively so that the review yields a defensible gap. The second is to

characterize the concrete limitations of existing clinical decision support, distinguishing genuine architectural shortfalls from problems that are merely unsolved engineering. The third is to develop a layered framework that integrates memory, reasoning, planning, RAG, and multi-agent collaboration as interacting layers rather than bolted-on features. The fourth is to design the specialized agents that populate that framework, with distinct responsibilities for monitoring, diagnostic support, risk prediction, treatment recommendation, and clinical explanation. The fifth is to embed trustworthy-AI principles, explainability, transparency, safety verification, auditability, and human-in-the-loop validation, into the architecture rather than treating them as an afterthought. The sixth is to ground the whole design in MIMIC-IV as the primary data source. The seventh, and the one that keeps the thesis honest about its limits, is to deliver a conceptual architecture together with a bounded prototype and a concrete evaluation plan that later work can implement and test at scale.

1.4 Research Questions

Five questions structure the investigation, and each maps onto a chapter of analysis rather than a single claim.

RQ1. What specific limitations of current LLM-based healthcare systems obstruct intelligent patient monitoring and clinical decision support, and which of them are architectural rather than incidental?

RQ2. How can Agentic AI improve reasoning, planning, memory management, and autonomous collaboration in a clinical setting without eroding physician control?

RQ3. How can multiple specialized agents collaborate on a clinical decision while preserving explainability and physician oversight at every step?

RQ4. How can Retrieval-Augmented Generation and external medical knowledge improve the reliability and transparency of AI-generated recommendations, and where does retrieval fail to help?

RQ5. How can the proposed framework support intelligent patient monitoring when instantiated on the MIMIC-IV dataset [14]?

The framing of RQ1 and RQ4 is intentional: each asks not only what a capability adds but where it stops working, because a review that only catalogs strengths cannot locate a real gap.

1.5 Scope of the Research

This work is a design study. Its deliverable is a conceptual Agentic AI framework, supported by a bounded prototype, rather than production software installed in a hospital. The framework integrates autonomous reasoning, memory management, RAG, planning, and multi-agent collaboration, and it is built around MIMIC-IV, which supplies anonymized critical-care records under a credentialed data-use agreement [14].

The design covers both structured and unstructured inputs, including demographics, admissions, laboratory results, vital signs, medications, diagnoses, procedures, and clinical notes. A clinician validation layer is part of the architecture by construction, so that recommendations assist rather than replace medical judgment.

Three things fall outside the scope, and naming them protects the claims that remain inside it. The thesis does not deliver production-ready hospital software, does not integrate with live EHR systems, and does not conduct a prospective clinical trial on real patients. What it does deliver is a framework specification and evaluation plan detailed enough to guide the implementation and validation that a follow-on project would undertake.

1.6 Significance of the Research

The contribution is integrative rather than atomic. Reasoning strategies, memory architectures, retrieval, and multi-agent coordination already exist as separate research threads, and each has been demonstrated in isolation [5][6][11]. The value here lies in specifying how they compose into a single clinical architecture and in making the seams between them explicit, because the seams are where most integrated systems fail. Layering trustworthy-AI mechanisms, explainability, transparency, auditability, and human oversight, into that composition is part of the same argument rather than a separate feature list.

For practice, the framework describes a system that watches patient state over time, retrieves relevant evidence, assesses risk, drafts recommendations, and explains each one in terms a clinician can check and overrule. Its stance toward the clinician is supportive by design: the human keeps final authority, and the architecture is arranged so that authority is easy to exercise rather than nominal.

The study also lays groundwork. By anchoring the design to MIMIC-IV and pairing it with an evaluation plan, it gives later researchers a concrete starting point for implementation and empirical testing rather than a diagram to reinterpret [14].

1.7 Research Methodology Overview

The methodology is design-oriented and proceeds in four phases. The first is a critical literature review spanning Agentic AI, LLMs, multi-agent systems, RAG, reasoning frameworks, and intelligent healthcare applications, read with the aim of exposing limitations rather than cataloging achievements [3][16].

The second phase converts that reading into a research gap through comparative analysis of recent systems, and organizes the findings as a taxonomy of LLM-based agents that classifies architectural components and research trends. The third phase proposes the layered framework itself, integrating MIMIC-IV data handling, memory management, reasoning modules, multi-agent orchestration, clinical decision support, and trustworthy-AI mechanisms; the full architecture is presented in Chapter 3. The fourth phase analyzes the proposed design conceptually and against existing approaches, and specifies how a bounded prototype would be evaluated, with the experimental design detailed in Chapter 4.

1.8 Thesis Organization

The thesis runs to five chapters. Chapter 1 sets out the background, problem, objectives, questions, scope, significance, and methodology. Chapter 2 reviews the literature on LLMs, Agentic AI, reasoning methods, multi-agent systems, RAG, healthcare AI, and trustworthy AI, presents the taxonomy of LLM-based agents, and identifies the research gap. Chapter 3 presents the proposed framework and methodology, describing the layered architecture, the MIMIC-IV data source, and the interaction among the agents. Chapter 4 sets out the experimental design and evaluation, including the setup, metrics, and planned analysis. Chapter 5 concludes by summarizing the contributions, revisiting the research questions, stating the limitations honestly, and outlining future work.

Note that the architecture is documented in Chapter 3, not Chapter 4; earlier draft labels that placed the framework figures in Chapter 4 are corrected throughout this revision.

1.9 Chapter Summary

This chapter argued that the obstacle to useful clinical AI is integration, not raw model capability. LLMs read and write medical language well but act as passive, memoryless responders, and existing decision support tends to be accurate yet narrow, opaque, and disconnected from current evidence. Agentic AI supplies the missing machinery, reasoning, planning, memory, tool use, and collaboration, but the healthcare systems built on it so far each solve only one part of the problem. The proposed framework answers that gap by composing these capabilities into a layered, supervised architecture grounded in MIMIC-IV. The next chapter reviews the underlying literature in depth, builds the taxonomy of LLM-based agents, and states the research gap that the framework is designed to close.

CHAPTER TWO

LITERATURE REVIEW

This chapter reviews the research that underpins the proposed Agentic AI framework and builds the argument for it critically rather than descriptively. It begins by tracing the move from Large Language Models to LLM-based agents and then to Agentic AI, examining the concepts, characteristics, and architectural layers that distinguish an agent from a passive model. It surveys the influential agent frameworks, ReAct, AutoGen, CAMEL, MetaGPT, and related systems, and organizes their capabilities into a taxonomy of memory, planning, reasoning, tool use, multi-agent collaboration, and healthcare application. The chapter then turns to the clinical domain, reviewing artificial intelligence and Large Language Models in healthcare, clinical decision support, Retrieval-Augmented Generation, and the trustworthy-AI mechanisms that any clinical system must satisfy. It closes with a research gap analysis that synthesizes these threads and a chapter summary. Throughout, the review reads prior work for what it fails to do as much as for what it achieves, because locating a defensible gap, not cataloguing achievements, is its purpose.

2.1 Introduction

This chapter presents a critical review of the literature on Agentic Artificial Intelligence (AI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), multi-agent systems, and their applications in healthcare. Its purpose is to establish the theoretical and technical foundation for the proposed Agentic AI framework for intelligent patient monitoring and clinical decision support, and, more importantly, to expose the limitations of existing work that the framework is designed to address.

Recent advances in Large Language Models have widened the reach of artificial intelligence well beyond conventional prediction. Modern systems reason, plan, manage memory, call external tools, and execute tasks with limited human intervention, and these capabilities have converged into Agentic AI, in which intelligent agents collaborate to solve problems through coordinated decision-making and interaction with external knowledge [3][4]. In healthcare, AI has produced promising results in diagnosis, clinical documentation, medical question answering, risk prediction, and treatment recommendation [1][16]. Persistent obstacles, hallucinated responses, opaque reasoning,

fragmented patient information, and the absence of continuous patient context, continue to slow clinical adoption, and researchers have proposed retrieval augmentation, multi-agent collaboration, persistent memory, and explainable AI to address them.

The review that follows does not merely summarize these contributions. It reads them for their architectural components, memory, reasoning, retrieval, orchestration, and trustworthy-AI safeguards, and it asks in each case what the cited work does not do. The chapter concludes with a research gap analysis that synthesizes these findings and motivates a unified framework combining reasoning, memory, Retrieval-Augmented Generation, multi-agent collaboration, and human-in-the-loop validation for clinical decision support grounded in the MIMIC-IV dataset [14].

2.2 Large Language Models and LLM-Based Agents

The arrival of Large Language Models changed how AI systems process information, generate language, and interact with users. Where earlier machine learning systems were trained for a single task on task-specific data, LLMs learn general competence from large-scale pre-training and can then perform language understanding, reasoning, summarization, question answering, and decision support without task-specific retraining [1]. That generality is the foundation on which everything in this chapter is built.

The limitation is equally foundational. A conventional LLM is a passive response generator: it receives a prompt, produces an answer, and terminates the interaction. However strong its language competence, it lacks the properties an autonomous system needs, persistent memory, long-term planning, interaction with an external environment, and independent decision-making. These gaps motivated the extension of LLMs into **LLM-based agents**, which wrap the model in additional components, memory systems, planning mechanisms, reasoning strategies, external tools, knowledge retrieval, and feedback loops, so that the system can perceive information, reason over it, act, and improve.

A typical LLM-based agent is organized around three modules. The **brain** is the reasoning capability supplied by the underlying language model, responsible for task understanding, decision-making, and intermediate reasoning. The **perception** module gathers information from external sources, databases, sensors, or user interaction. The **action** module executes tasks through tools, APIs, software systems, or interaction with

other agents. This decomposition is what turns a text generator into something that can pursue a goal over several steps.

Several lines of work supplied the machinery that makes such agents effective. The ReAct pattern interleaves reasoning traces with task-specific actions, letting an agent analyze a problem, act on the environment, observe the result, and revise its plan, which improves performance on tasks that require both reasoning and external retrieval [5]. Generative Agents introduced a memory-stream architecture in which an agent records experiences and observations, retrieves relevant history, reflects, and plans, a capability directly relevant to healthcare, where previous diagnoses, clinical history, treatment responses, and monitoring observations must persist over long periods [8]. Multi-agent frameworks such as AutoGen, CAMEL, and MetaGPT extended the paradigm further, showing how specialized agents with distinct roles can communicate and coordinate to solve problems no single agent handles well, an arrangement that mirrors the collaboration among specialists in clinical care [6][7][18]. A further advance was the integration of Retrieval-Augmented Generation, which connects a model to external knowledge, medical databases, guidelines, literature, and records, so that responses rest on retrieved evidence rather than parameters alone, reducing hallucination and improving reliability [15].

In healthcare these ideas have produced early but instructive systems. MedAgents stages collaborative medical reasoning among specialist agents, MedRAG couples retrieval with structured medical knowledge, Agent Hospital simulates an autonomous care environment, and Med-PaLM and related models demonstrate strong medical question answering [10][11][12][2]. Each shows that LLM capabilities can be adapted to medical reasoning, yet each also inherits the reliability, transparency, and privacy concerns that make clinical deployment hard: hallucination, biased recommendations, faulty reasoning chains, and limited explainability remain unresolved. This is why trustworthy-AI mechanisms, human-in-the-loop validation, explainable recommendations, audit logging, and safety verification, are not optional additions but preconditions for any healthcare-focused agent, a theme developed in Section 2.9. For the present research, LLM-based agents supply the foundation on which memory, reasoning, planning, retrieval, and specialized clinical agents are composed into the proposed framework.

2.3 Agentic AI: Concepts and Evolution

2.3.1 From Fluent Text to Goal-Directed Action

Large Language Models made text generation cheap and fluent, and in doing so they exposed how little fluency alone accomplishes when a task requires acting over time. Out of that gap grew Agentic AI: systems that do not merely produce text but analyze a situation, form a plan, take actions, and adjust when the environment answers back [3][4]. The distinction is easy to blur, and Sapkota and colleagues take pains to separate a single tool-using model from a coordinated collection of specialized agents, arguing that only the latter earns the label "agentic" [9]. That separation is not pedantic. A monitoring task that spans a multi-day admission cannot be served by a model that forgets each exchange, so the architectural difference has direct clinical consequences.

Where a task-specific model is trained to do one thing, an agent is organized around a goal it decomposes on its own. It reads a complex objective, breaks it into steps, chooses a strategy, reaches for external tools when its own knowledge runs out, and coordinates with other agents when the problem exceeds any single role [4]. In healthcare this proactive stance matters because clinical work is itself a loop of observation, hypothesis, evidence-gathering, and revision rather than a one-shot answer. The critical caveat, easy to lose in the enthusiasm, is that autonomy is a liability as much as a feature in medicine: the same initiative that lets an agent chase down evidence also lets it compound an early error across several steps before anyone notices, which is why the sections that follow treat oversight as a design constraint rather than an add-on.

2.3.2 Evolution from Traditional AI to Agentic AI

The path to Agentic AI runs through several stages, and each stage solved a problem while leaving another untouched. Early clinical AI was rule-based: expert systems encoded medical knowledge as explicit if-then logic. They worked inside narrow, well-specified scenarios, but they demanded painstaking manual knowledge engineering and broke whenever a guideline changed or a patient failed to match a coded pattern. Because real medicine is thick with uncertainty and interacting factors, rule-based systems could never cover the space they were meant to serve.

Machine learning shifted the burden from hand-written rules to learned patterns, which enabled risk prediction, readmission estimation, and early-warning scores drawn

from historical data. Deep learning pushed further, extracting features automatically from notes, images, and physiological signals. What neither delivered was reasoning a clinician could follow or interaction that extended beyond a fixed prediction; the output was a number, and the justification stayed inside the network. LLMs then loosened the format constraint by handling open-ended language, but the earliest ones were conversational engines with no persistent memory, no planning, and no way to touch the world outside their context window. Agentic AI closes these gaps by wrapping an LLM in additional machinery, memory across short and long horizons, planning and task decomposition, explicit reasoning, tool use, Retrieval-Augmented Generation, multi-agent collaboration, and self-reflection [5][19][15]. The important point is that these components are not merely accumulated; they convert a response generator into a system that can pursue a goal. That said, each added component is also an added failure mode, and the literature has been quicker to demonstrate the capabilities than to characterize how they fail together.

2.3.3 Characteristics of Agentic AI Systems

Several properties separate an agentic system from a conventional model, and a clinical framework has to earn each one. **Autonomous decision-making** lets an agent read a goal, weigh the available information, and choose an action from its own reasoning [4]; in monitoring this could surface an abnormal trend or draft a recommendation before a clinician asks, but unsupervised autonomy in a safety-critical setting can act on a wrong inference as readily as a right one, so its value is bounded by how tightly it is coupled to review. **Goal-oriented planning** lets an agent split a complex objective into an ordered set of steps, analyzing vital signs, retrieving history, comparing against guidelines, assessing risk, and only then drafting a recommendation, yet a plan is only as sound as the model's grasp of the domain, and a confidently wrong plan is harder to catch than a single wrong answer.

Memory and context awareness is the property most obviously missing from a plain LLM exchange. Agentic systems keep short-term conversational state, long-term patient history, and vector-based semantic stores so that later decisions draw on earlier context, which for longitudinal monitoring is not optional; the open question, largely unresolved in the current literature, is how to keep such memory both clinically faithful and privacy-compliant as it grows. **Reasoning and self-reflection** interpose deliberation

between observation and action: the ReAct pattern interleaves reasoning with tool calls [5], reflection mechanisms such as Reflexion let an agent critique and revise a prior attempt [19], and structured reasoning methods such as Chain-of-Thought, Self-Consistency, and Tree of Thoughts improve multi-step inference [20][21][22]. Self-reflection is valuable where a wrong recommendation carries real cost, but it also risks rationalizing an initial error rather than correcting it, so the framework treats it as support for human review rather than a substitute. **Tool utilization** extends an agent past its internal knowledge through external systems and APIs, an ability formalized in Toolformer [23] and demonstrated at scale in embodied settings such as Voyager [24]; tool use grounds an agent in current information but imports the reliability and latency of whatever it calls. **Multi-agent collaboration** divides a problem among specialized agents that communicate to reach a joint result, as in AutoGen and CAMEL [6][7]; division of labor mirrors how clinical teams work but multiplies the communication surface, and errors that pass silently between agents are among the least studied risks in this line of research.

2.3.4 Agentic AI Architecture

A typical agentic architecture stacks several cooperating layers, and mapping them onto healthcare clarifies both the fit and the strain. A perception layer ingests information from the environment, which in a clinical system means EHR records, demographics, laboratory results, vital signs, notes, and medication history. A memory layer stores that information and maintains context, combining short- and long-term stores with vector databases and knowledge repositories so that agents can recall relevant prior experience. A reasoning layer analyzes what perception and memory supply, drawing on Chain-of-Thought prompting, ReAct-style reasoning, and retrieval-augmented inference to reach a decision [20][5][15]. A planning and orchestration layer sequences tasks and routes them among agents, with a coordinator deciding which agent acts and when. An action layer carries decisions back out, generating recommendations, updating records, or communicating results.

Presented this cleanly, the layered picture can imply more maturity than exists. In practice the boundaries between reasoning, memory, and orchestration are contested, and most published architectures specify the layers far more precisely than they specify how a layer fails when it returns something wrong. The framework proposed in this thesis

inherits that risk and addresses it by placing verification and human review across the orchestration and action layers rather than trusting the stack to be self-correcting; the full architecture is presented in Chapter 3.

2.3.5 Agentic AI in Healthcare

Healthcare is a natural target for agentic methods because medical decisions demand exactly what agents supply: continuous observation, multi-step reasoning, and the fusion of diverse information. A cluster of recent systems shows both the promise and the limits. MedAgents organizes a role-played consultation among specialist agents and reports gains on medical reasoning benchmarks, though its evaluation centers on question answering rather than on monitoring a patient over time [10]. MedRAG fuses retrieval with structured medical knowledge to improve diagnostic specificity, which helps most when candidate diagnoses share symptoms, yet it remains a diagnostic reasoning system rather than a full workflow [11]. Agent Hospital simulates a whole care environment populated by autonomous agents and is valuable as a testbed, but a simulation of patients is not evidence about real ones [12]. Clinical Camel adapts an open model for medical dialogue and lowers the barrier to healthcare-specific LLMs while inheriting the memory and planning limits of the underlying model [13]. Read together, these systems make a consistent point: each demonstrates one capability convincingly and stops short of integrating the rest.

2.3.6 Challenges and Limitations of Agentic AI

Several obstacles stand between these demonstrations and clinical deployment, and none is fully solved. Reliability is the first: agents can generate unsupported recommendations, and in medicine such errors reach the patient; retrieval grounding reduces but does not eliminate the problem, because a confidently reasoned chain can still rest on a mis-retrieved fact [11]. Explainability is the second, because clinicians will not accept a black-box recommendation, so an agent that cannot show its evidence and reasoning is, for practical purposes, unusable regardless of accuracy. Data privacy is the third, since agentic systems accumulate sensitive patient information in memory and logs, and secure, compliant handling is a precondition rather than a nicety. Safety and alignment form the fourth: autonomous agents must stay inside clinical guidelines and ethical bounds, which is why human oversight remains necessary for consequential decisions. The fifth is cost, since memory, reasoning, retrieval, and multiple coordinated

agents together impose a computational and latency burden that a real-time monitoring setting can ill afford. The honest summary is that the research literature has demonstrated capability far more thoroughly than it has demonstrated safety, and a clinical framework has to make up that difference deliberately. Agentic AI accordingly supplies the theoretical foundation for the proposed framework, which adopts these principles concretely, processing MIMIC-IV records, managing patient context, retrieving knowledge through RAG, coordinating specialized agents, and routing every recommendation through human validation, while remaining accountable to the clinician who signs off on its output [14][15].

2.4 Agentic AI Frameworks

The development of LLM-based agents has produced a set of general-purpose frameworks that each contribute a distinct capability toward intelligent systems. An agentic architecture generally combines a memory module that maintains historical information, a reasoning engine for logical analysis, a planning module that decomposes objectives into executable tasks, a tool-integration layer that reaches external resources, a multi-agent communication mechanism, and a knowledge-retrieval component built on techniques such as RAG. The frameworks reviewed below realize different subsets of these capabilities, and their strengths and omissions together define what a healthcare-specific framework must supply.

The **ReAct** framework marked an important shift by combining reasoning with external action. Rather than answering from internal knowledge alone, a ReAct agent reasons about the current situation, selects an action, receives an observation from the environment, and updates its reasoning before continuing, iterating until it reaches a decision [5]. This suits clinical decision support, where a diagnosis is refined as new evidence arrives: an agent can analyze symptoms, retrieve guidelines, weigh candidate diagnoses, and revise its recommendation. Its principal limitation is memory: ReAct depends on the current context window and offers no dedicated long-term store, which makes it awkward for continuous monitoring across an extended admission.

The **AutoGen** framework provides a flexible architecture for applications built from multiple interacting agents [6]. Instead of a single agent, AutoGen composes specialized agents that communicate through conversation, decompose tasks, execute external tools, and admit human-in-the-loop interaction. In a clinical configuration a

monitoring agent could analyze vitals, a diagnosis agent evaluate conditions, a treatment agent suggest interventions, and a verification agent check recommendations against guidelines. Its strength is scalable multi-agent coordination; its open problems are communication efficiency, computational cost, and reliable coordination among autonomous agents. The **CAMEL** framework focuses on autonomous communication and cooperation through role-based collaboration, assigning each agent a defined role, planner, domain expert, critic, or executor, so that agents cooperate with limited human guidance [7]. Role-playing reduces ambiguity and organizes collaboration, and it provides a natural template for specialized clinical agents, but CAMEL still struggles with communication loops, the propagation of incorrect reasoning, and drift from human objectives.

The **MetaGPT** framework introduces structured multi-agent collaboration by importing software-engineering practice, predefined roles, shared information management, structured workflows, and Standard Operating Procedures (SOPs) [18]. Controlled, SOP-driven workflows are especially relevant to healthcare, where practice follows established protocols, and a MetaGPT-inspired clinical pipeline could move from patient-data analysis through assessment, diagnosis, treatment recommendation, and safety verification. The trade-off is adaptability: SOP-bound workflows need further work to support continuous learning in dynamic clinical settings. The **Agent Hospital** framework is one of the earliest attempts to apply agentic concepts directly to healthcare, placing autonomous medical agents in a simulated hospital where they consult, diagnose, plan treatment, and improve by accumulating medical-case memory and learning from successful and failed outcomes [12]. Its learning-from-experience design is highly relevant to monitoring, which likewise depends on patient history and outcomes, but its limitations are real: little real-world clinical validation, no multimodal data integration, and a need for stronger safety mechanisms.

Beyond these, several general-agent systems fill in specific capabilities the clinical setting will need. **Generative Agents** contributed the memory-stream-and-reflection design that lets an agent maintain coherent long-horizon behavior [8]. **Reflexion** added verbal self-reflection, in which an agent critiques a failed attempt and stores the critique to improve subsequent ones [19]. **Toolformer** showed that a model can learn, in a self-supervised way, when and how to call external tools, improving factual accuracy [23], while **Voyager** demonstrated open-ended skill acquisition and tool use by

an embodied agent in a complex environment [24]. None of these targets medicine, but each supplies a mechanism, durable memory, reflective revision, disciplined tool use, that a clinical framework has to incorporate and constrain.

A comparative view makes the pattern plain. When these frameworks are scored on general capabilities, reasoning, communication, role-based collaboration, workflow control, and healthcare orientation, they distribute their strengths unevenly and none covers the full set required for supervised longitudinal monitoring.

[Table 2.1: Comparative analysis of reviewed frameworks (insert table here)]

The shared shortcoming is decisive for this thesis. Existing frameworks target general-purpose task completion, not healthcare-specific decision support, and they provide limited long-term patient memory, weak integration with real clinical datasets, no continuous monitoring, limited explainability of autonomous decisions, and no built-in human-in-the-loop clinical validation. The proposed research responds by integrating MIMIC-IV data, patient memory management, Retrieval-Augmented Generation, clinical reasoning agents, multi-agent collaboration, trustworthy-AI mechanisms, and human validation into a single framework, bridging the distance between general-purpose LLM agents and reliable healthcare decision support.

2.5 Taxonomy of LLM-Based Agents

Because LLM-based agents differ widely in capability, architecture, and application, it helps to classify them along the functional dimensions that recur across the literature. A workable taxonomy organizes agents by six capabilities, memory, planning, reasoning, tool use, multi-agent collaboration, and healthcare application, and reading existing systems against these dimensions clarifies what an intelligent healthcare agent must combine and where prior work stops short [4].

Memory is the capability that lets an agent maintain context beyond a single input window, spanning short-term, long-term, semantic, episodic, and vector-based forms. Generative Agents illustrated the value of a memory stream that records experiences, retrieves relevant history, and supports reflection [8], and surveys of autonomous agents consistently identify memory as central to autonomy, personalization, and decision consistency [4]. For clinical use, memory is what converts isolated analysis of single events into continuous patient understanding across previous diagnoses, medication

history, treatment responses, and observations. **Planning** transforms high-level goals into executable action sequences through task decomposition, hierarchical planning, feedback-driven revision, and multi-step reasoning; ReAct exemplifies the interleaving of reasoning and action, and AutoGen and MetaGPT show how planning distributes across collaborating agents under conversational or SOP-driven control [5][6][18].

Reasoning lets an agent analyze complex problems and weigh evidence, which clinical decisions demand because they depend on history, symptoms, laboratory results, guidelines, and prior outcomes at once. ReAct improves transparency by exposing intermediate reasoning [5], MedAgents shows that collaborative reasoning among simulated experts can raise diagnostic accuracy and reduce individual-model error [10], and MedRAG combines retrieval with medical knowledge to ground reasoning in evidence [11]. **Tool use** addresses the fixed knowledge boundary of a pre-trained model by connecting it to search engines, databases, APIs, and knowledge repositories; Toolformer showed a model can learn to call tools to improve factual accuracy [23], and RAG has become the most important tool for healthcare agents because it lets them retrieve relevant evidence, drug information, guidelines, and records, before generating a response [15]. **Multi-agent collaboration** distributes responsibility across specialized agents: CAMEL structures role-based communication [7], AutoGen enables flexible multi-agent conversations with tool execution and human involvement [6], MetaGPT enforces role-based workflows through SOPs [18], and Agent Hospital demonstrates an autonomous healthcare ecosystem of interacting agents [12].

The **healthcare application** dimension gathers the systems that adapt these capabilities to medicine, MedAgents for collaborative reasoning, MedRAG for retrieval-grounded diagnosis, Med-PaLM for medical question answering, Clinical Camel for privacy-aware open medical dialogue, Agent Hospital for simulated autonomous care, and Med-PaLM M for multimodal biomedical intelligence, each strong on one or two dimensions and weak on the others [10][11][2][13][12][25]. The proposed framework, presented in Chapter 3, is deliberately positioned to span all six dimensions at once: longitudinal patient memory, a task planner and workflow coordinator, a ReAct-based clinical reasoning engine, RAG-based medical knowledge retrieval, specialized collaborating healthcare agents, and a patient-monitoring and decision-support application layer. Reading the taxonomy this way shows that the individual capabilities

are established while their integration under clinical supervision is not, which is the space the framework occupies.

2.6 Artificial Intelligence in Healthcare

Artificial intelligence has become a transformative technology in healthcare by enabling systems to perform tasks that traditionally require human intelligence, pattern recognition, decision-making, prediction, and knowledge discovery, across the diverse and complex datasets medicine produces: Electronic Health Records, medical imaging, laboratory results, physiological signals, genomic information, and clinical narratives [1]. The growth of digital health data, computational capacity, and machine learning has accelerated adoption, but progress has been uneven, and understanding where AI succeeds and where it stalls frames the need for the proposed work.

2.6.1 Evolution and Applications

Healthcare AI has passed through the same stages traced earlier in this chapter. Rule-based expert systems encoded medical knowledge manually and assisted with specific tasks but lacked adaptability. Machine learning shifted toward data-driven pattern recognition, enabling disease classification, risk assessment, and outcome prediction while depending on engineered features and large structured datasets. Deep learning expanded capability in complex-representation tasks, with convolutional networks succeeding in radiological image analysis and transformer architectures improving sequential clinical data processing. Most recently, foundation models and Large Language Models introduced a phase in which one model performs many language-based healthcare tasks, clinical summarization, medical question answering, knowledge retrieval, and patient communication, though limits in autonomy, reliability, and continuous interaction motivated the move toward Agentic AI [1][16].

These capabilities have been applied across the clinical workflow. In diagnosis and prediction, models analyze patient characteristics, history, laboratory values, and imaging to identify disease patterns across cardiovascular, oncological, neurological, and infectious conditions, though such systems often focus on a single disease or dataset and support comprehensive decision-making poorly. In medical imaging, deep learning detects abnormalities, segments structures, and triages critical cases, improving efficiency but typically operating in isolation from history, laboratory results, and medications. In

patient monitoring and risk prediction, machine learning models estimate deterioration, mortality, and readmission risk from real-time and historical data, yet frequently deliver predictions without the explanations clinicians need before acting. In clinical decision support, systems provide evidence-based recommendations and alerts but, when built on predefined rules or statistical models, struggle with incomplete information, changing conditions, and interacting factors, precisely the situations where LLMs and Agentic AI create new room to reason over clinical information, retrieve knowledge, and coordinate specialized agents.

2.6.2 Large Language Models in Healthcare

General-purpose language models perform impressively, but healthcare is a high-risk domain where accuracy, explainability, and patient safety are essential, which has driven the development of medical LLMs. **Med-PaLM** was among the earliest, adapting Google's PaLM architecture for medical question answering and evaluating on the MultiMedQA benchmark, using instruction prompt tuning and physician assessment of factual accuracy and potential harm to improve clinical reasoning [2]. It relies on internal model knowledge, however, and includes neither persistent patient memory nor external retrieval. **Med-PaLM 2** extended it with an ensemble-refinement strategy that generates multiple reasoning paths before selecting a response, improving diagnostic reasoning and benchmark performance and reducing harmful outputs, yet it too depends largely on pre-trained knowledge and lacks continuous integration with patient-specific information [26]. **Med-PaLM M** addressed the single-modality constraint by introducing a generalist biomedical model that processes medical images, clinical text, and biomedical reasoning within one architecture, which is valuable for monitoring decisions that combine structured and unstructured data [25]. **Clinical Camel** took a different path, offering an open-source medical model trained through dialogue-based knowledge encoding to support transparent, reproducible, and privacy-aware healthcare applications, demonstrating that open models can be competitive [13]. More recent conversational diagnostic systems such as AMIE push toward multi-turn diagnostic dialogue, extending the reach of medical LLMs while remaining research prototypes [27].

Across these systems, applications extend well beyond question answering to clinical documentation, report generation, diagnosis, treatment recommendation, summarization, drug-information retrieval, patient education, and workflow automation,

with real potential to reduce documentation burden and support complex decisions [16]. The limitations are equally consistent, and they define the gap. Models still hallucinate clinically incorrect information with high confidence; most maintain no persistent long-term patient memory across encounters; many cannot access updated guidelines or new evidence at inference; explanations are often thin where clinicians require transparent reasoning; and validation is frequently confined to benchmark datasets rather than real hospital environments. In short, current medical LLMs function as standalone models rather than intelligent healthcare agents, with limited integration into autonomous multi-agent architectures, no continuous monitoring, absent persistent memory, limited real-time retrieval, weak collaboration among specialized agents, and insufficient trustworthy-AI mechanisms. These shortcomings are exactly what the proposed framework sets out to address by pairing medical LLMs with retrieval, persistent memory, specialized agents, and human oversight, grounded in MIMIC-IV [14].

2.7 Clinical Decision Support

Clinical Decision Support Systems (CDSS) analyze patient information and return knowledge, alerts, recommendations, or insights that help a clinician decide. They are among the oldest applications of computing in medicine, and their purpose has stayed constant even as their methods have not: improve care quality, cut preventable errors, and keep practice tied to evidence. The pressure behind them has only grown, because a clinician now has to hold patient history, laboratory values, medications, active diagnoses, guidelines, and prior outcomes in view at once, and in critical care that synthesis has to happen fast. Manual integration of everything relevant is often simply not feasible, which is the standing motivation for decision support. Traditional CDSS delivered real value, but limits in adaptability, scalability, and contextual understanding pushed researchers toward AI-driven designs.

2.7.1 Evolution and Components

CDSS has moved through generations, and each traded one weakness for another. The earliest systems were expert systems, with medical knowledge hand-encoded as rules so that a defined condition triggered a defined recommendation; they performed well in narrow scenarios but demanded heavy manual knowledge engineering and grew brittle when a guideline changed or a patient failed to fit a coded pattern. Machine learning changed the sourcing of knowledge from authored rules to learned patterns, opening up

disease prediction, mortality-risk estimation, readmission prediction, and early-warning scores, and deep learning extended the reach by extracting features from notes, images, and signals. The persistent shortfall across both is that most such systems output a risk score or classification without reasoning, explanation, or dialogue; a number arrives, the account of why does not, and that missing account has kept much of this work at the edge of routine practice.

A modern CDSS is usually built from four cooperating parts, each carrying its own difficulty. A data-acquisition layer gathers information from EHRs, laboratory systems, monitoring devices, and documentation, and because healthcare data mixes structured values with unstructured text, it must reconcile formats never designed to be read together, so incomplete integration yields an incomplete picture of the patient. A knowledge base holds the medical information the system reasons over, drawn increasingly from literature, guidelines, biomedical databases, and knowledge graphs; recommendations are only as trustworthy as this base, and keeping it current is a never-finished burden. An inference and decision engine forms the reasoning core, with recent LLM-based approaches adding the ability to interpret clinical narratives and produce natural-language explanations, though the added flexibility brings added opacity [1]. A user interface mediates between system and clinician and decides whether a technically sound system is actually used, since an interface that adds cognitive load defeats its purpose, and a good one presents not only a recommendation but the evidence, reasoning, and confidence behind it.

2.7.2 Limitations and the Role of Agentic AI

Several limitations remain unresolved, and naming them precisely defines the target for anything claiming to improve on the state of the art. Limited context understanding leaves many systems reasoning over a handful of predefined variables rather than the whole patient, so they can be confidently incomplete. Limited adaptability forces manual updating when evidence changes, leaving recommendations trailing the literature, which motivates the retrieval mechanisms discussed in Section 2.8. Limited explainability leaves many AI systems, deep learning ones especially, operating as black boxes that may be accurate yet unable to say why, and in medicine a recommendation a clinician cannot interrogate is one a clinician should not accept. Fragmented intelligence means most platforms handle a single task without coordinating across tasks, which maps

poorly onto genuinely collaborative, multi-specialty clinical reasoning. Finally, the human-AI interaction problem is that consequential decisions require human judgment and accountability, so fully autonomous decision-making introduces a safety risk no accuracy figure offsets; the defensible posture is human-in-the-loop, with the system supplying recommendations, evidence, and explanations while the clinician retains final authority.

Recent AI has opened room to address these limitations. LLMs process clinical text, summarize records, and support medical reasoning, and healthcare-tuned models such as Med-PaLM and Clinical Camel show competence in medical question answering and clinical knowledge representation [2][13]. Multi-agent designs such as MedAgents stage collaboration among specialized agents to approximate an expert consultation [10], and Retrieval-Augmented Generation lets a model draw on external medical sources during inference, reducing hallucination and improving reliability [15][11]. These gains are real but partial: a standalone LLM-based CDSS still lacks durable memory, cross-step planning, and continuous monitoring, so it improves the quality of an answer without changing the shape of the workflow. That residual gap, capable reasoning without an architecture to sustain it over time, points toward integrating LLMs with Agentic AI, in which a monitoring agent tracks patient state, a risk-prediction agent watches for complications, a diagnosis agent weighs candidate conditions, a treatment agent proposes interventions, and an explanation agent renders the reasoning legible, all coordinated by an orchestrating agent that draws on memory, guidelines, and external knowledge [3][4]. This decomposition mirrors how clinical teams distribute expertise and directly attacks fragmentation, but dividing work among agents multiplies the interfaces where an error can pass unnoticed, so the architectural benefit is real only if coordination and verification are engineered as carefully as the agents themselves.

2.8 Retrieval-Augmented Generation in Healthcare

Large Language Models read and write medical language fluently, yet fluency masks three failure modes that are especially damaging in clinical use: their knowledge is frozen at training time, they hallucinate plausible falsehoods, and they cannot reach patient-specific information on their own. Clinical decisions need recommendations that are both accurate and backed by current, verifiable evidence, and a model reasoning only from its internal parameters cannot reliably meet that bar [1]. Retrieval-Augmented

Generation was designed to close this shortfall. RAG separates knowing from remembering: rather than answering from parameters alone, the model first retrieves relevant material from an external source and then generates a response conditioned on it, so the answer rests on evidence that can be inspected [15][17]. In healthcare that source can be EHRs, clinical guidelines, the medical literature, drug databases, or a patient's own prior records. The approach is powerful, but retrieval solves the knowledge-access problem without solving the trust problem, and conflating the two is a common error in the healthcare RAG literature.

2.8.1 Concept and Importance

RAG combines a retriever, which searches external knowledge for material relevant to the query, with a generator, which conditions its response on that material together with the query [15]. The pipeline moves from a user query or patient information, through information retrieval, to a relevant knowledge context, into the language model, and finally to a generated response. In a clinical setting the retrieved context might include records, guidelines, research articles, drug-information entries, and prior diagnoses and treatments. Grounding a response this way lowers the chance of an unsupported recommendation, but only to the degree the retrieval itself is sound; a fluent answer built on an irrelevant or outdated passage is no safer for having been retrieved, which makes retrieval quality a first-order concern rather than a detail.

Three benefits make RAG important for medicine, each with a caveat. Knowledge updating lets a system retrieve current material at inference, so responses reflect present evidence rather than the state of the world at training time, though this only helps if the underlying repository is itself kept current, relocating rather than removing the maintenance burden [17]. Reduction of hallucination follows from anchoring reasoning to authoritative sources, but a model can still misread, over-generalize, or selectively use a correctly retrieved passage, so retrieval lowers the hallucination rate without driving it to zero [15]. Personalized decision support comes from pulling patient-specific material, demographics, laboratory results, vital signs, prior diagnoses, medications, and notes, so that reasoning is tailored rather than generic, which is precisely what makes RAG valuable for continuous monitoring, where the relevant context is the specific patient's evolving record. More advanced variants such as Self-RAG add self-reflective control

over when to retrieve and whether retrieved evidence actually supports the generated claim, a step toward the verification the clinical setting demands [28].

2.8.2 Retrieval in Agentic Systems and MedRAG

Inside an agentic framework, RAG is the primary mechanism by which agents acquire knowledge: an agent retrieves evidence before committing to a decision rather than reasoning in a vacuum. In a clinical configuration the retrieval task differs by role. A monitoring agent retrieves previous vital-sign trends and patient history; a diagnosis agent retrieves similar cases, candidate diseases, and clinical guidelines; a risk-prediction agent retrieves risk-assessment models and historical outcomes; a treatment agent retrieves medication guidelines and protocols; and an explanation agent retrieves the supporting evidence for a recommendation. Coordinating retrieval this way lets several agents reason from a shared evidentiary basis, an advantage over independent lookups, but the accompanying risk, largely unaddressed in existing multi-agent RAG work, is that a single poor retrieval can propagate through every agent that consumes it, so retrieval quality and the human-validation step must safeguard the whole chain rather than any one agent.

MedRAG illustrates the state of the art and its limits. It extends conventional RAG by folding structured medical knowledge into reasoning, combining clinical knowledge graphs, retrieval, diagnostic reasoning, proactive patient questioning, and evidence-based recommendation, and pairing structured knowledge with retrieval improves diagnostic specificity when diseases present with overlapping symptoms, exactly where unstructured retrieval blurs candidates together [11]. Its boundary is scope: MedRAG is a diagnostic reasoning system evaluated largely on question-answering-style tasks, so it strengthens one stage of care rather than sustaining a monitoring workflow over time, which is part of what the proposed framework extends.

2.8.3 RAG with MIMIC-IV and Open Challenges

The proposed research uses MIMIC-IV as its primary clinical data source [14]. The dataset holds comprehensive critical-care records, demographics, admissions, ICU stays, laboratory measurements, vital signs, notes, medications, procedures, and diagnoses, which together give the framework enough longitudinal detail to reason about a patient over an admission rather than at a single instant; it builds on the broader movement toward openly available physiological and clinical data resources for research

[29]. Within the framework, records are processed into vector embeddings stored in a vector database, and at inference the retriever performs a semantic search for material relevant to the current patient's condition, surfacing similar prior cases and relevant detail to support context-aware recommendations. One caveat bounds every claim built on this data: MIMIC-IV is retrospective, single-source ICU data, so retrieval over it reflects the practice patterns of one institution at one time and cannot stand in for prospective, multi-site evidence [14]. The framework uses MIMIC-IV to design and demonstrate the approach, not to establish clinical generalizability.

Several challenges remain open even where RAG helps. Data privacy is the most immediate, since healthcare information is highly sensitive and secure storage, controlled access, and anonymization are preconditions for deployment. Retrieval quality is the most consequential for output, because a recommendation inherits the relevance of whatever was retrieved, and poor retrieval quietly injects misleading evidence into otherwise sound reasoning. Knowledge maintenance is the most persistent, since a repository that is not actively curated decays into the same staleness RAG was meant to fix. Explainability is the most decision-critical, because clinicians need to see the evidence, confidence, and references behind a recommendation, and a system that retrieves silently offers little more trust than the bare model it replaced. Taken together these show that retrieval is a necessary component of trustworthy clinical AI but not a sufficient one, which is why most RAG-based healthcare systems, having advanced medical question answering and evidence retrieval, still stop at isolated retrieval rather than supporting a full clinical workflow [11][17]. This research embeds RAG inside a larger Agentic AI framework, coupling retrieval with autonomous clinical agents, persistent patient memory, reasoning modules, and human oversight, drawing its knowledge from MIMIC-IV, so that evidence retrieval and multi-agent collaboration together deliver decision support that is reliable, explainable, and patient-centered [14][15].

2.9 Trustworthy AI in Clinical Decision Support

Accuracy is necessary for clinical AI but nowhere near sufficient. Because a recommendation can change how a patient is treated, the system that produces it has to be reliable, transparent, and answerable in ways a diagnostic classifier never had to be. This is the demand the notion of Trustworthy AI tries to capture: artificial intelligence that is safe, explainable, fair, and accountable rather than merely performant [30]. In medicine

the stakes give the word "trust" an operational meaning, since a physician who cannot see why a recommendation was made cannot responsibly act on it, and adoption stalls exactly there. The mechanisms that turn trustworthiness from an aspiration into architecture, explainability, safety, fairness, auditability, and human oversight, are examined below, and most current systems name these goals more thoroughly than they implement them.

Explainable AI answers the oldest problem, that deep networks and LLMs predict well while offering no legible account of why, by surfacing the evidence and reasoning behind an output so a clinician receives a justification alongside a recommendation [30]. Explanation raises confidence, supports independent validation, makes errors easier to spot, eases regulatory review, and gives clinicians language to share with patients. Within the proposed framework this responsibility falls to a dedicated explanation agent that summarizes the reasoning path and names the evidence retrieved through RAG [15]; the critical limitation, which the framework cannot fully escape, is that a fluent explanation is not proof of a faithful one, since a model can narrate a plausible rationale that does not correspond to the computation that produced the recommendation, so explanation supports human review without replacing it.

AI safety places patient safety above every other objective, so a clinical system must actively minimize the chance of an incorrect, harmful, or misleading recommendation rather than merely maximize average accuracy, through mechanisms that check recommendations against guidelines, detect contradictions, estimate confidence, flag uncertain predictions, and cross-validate across reasoning paths. These are guardrails around an assistant, not a basis for autonomy: guideline checks and confidence estimates catch the failures they were designed to anticipate and are blind to novel ones, so safety tooling reduces risk without eliminating it, and the human-review step exists precisely to cover what the tooling misses. Bias and fairness matter because clinical datasets encode the disparities of the care that generated them, imbalanced samples, missing information, demographic underrepresentation, and historical inequities, which a model can learn and reproduce; fairness monitoring evaluates performance across subgroups to detect such gaps, but bias in medicine is not a single quantity, since statistical parity, equal error rates, and calibration across groups can conflict, so the framework commits to measuring and reporting subgroup performance transparently rather than claiming any procedure removes bias.

Auditability and accountability require a record: a clinical system should log which data it used, which evidence it retrieved, which reasoning steps it took, how the agents interacted, what it proposed, and how the human responded, so that a clinician or auditor can reconstruct how a recommendation arose. Such logs underpin transparency, quality assurance, and compliance, and they make accountability real when a decision is later questioned, but a log is only useful if it is complete and tamper-evident, and one that records outputs without the retrieved evidence or the human's final action leaves exactly the gaps an audit needs to close. Human-in-the-loop validation ties these together: efficient analysis of large data volumes does not confer clinical authority, so the physician reviews and approves every AI-generated recommendation, and the workflow moves from patient data, through the agentic framework, to a clinical recommendation, to physician review, to an explicit approve, modify, or reject decision, and only then to a final clinical action. This pairs computational throughput with clinical judgment, but its documented limitation is automation bias, the tendency of reviewers to defer to a confident machine suggestion, so the review step protects patients only if the interface presents evidence in a way that invites genuine scrutiny, which is partly why the framework emphasizes explanation and audit logging.

The proposed framework accordingly carries a dedicated trustworthy-AI layer so these safeguards are structural rather than optional, bringing together an explainability module, a safety-verification module, a bias-monitoring module, an audit-logging system, and human-in-the-loop validation, running alongside the reasoning and orchestration layers so that no recommendation reaches a clinician without supporting evidence and a record of how it was produced [30]. Placing trust mechanisms in their own layer is itself a claim, that trustworthiness is a first-class architectural concern, and the burden the thesis accepts is to show these components interacting rather than merely listed. This matters because, although trustworthy AI is now a stated priority, the gap between statement and implementation is wide: many systems report strong predictive performance while lacking coherent mechanisms for transparency, accountability, and continuous monitoring, and most agentic frameworks are built for general-purpose use and never confront the specific ethical and regulatory demands of clinical settings [30]. The framework's response is to integrate these safeguards into the architecture rather than append them, while remaining honest that these mechanisms manage risk rather than eliminate it.

2.10 Research Gap Analysis

The literature reviewed in this chapter shows substantial progress across Large Language Models, Agentic AI, Retrieval-Augmented Generation, and AI-driven clinical decision support. Intelligent agent architectures reason, plan, use tools, and collaborate [5][6][18], and healthcare-focused research has shown LLMs assisting with medical question answering, diagnosis support, and documentation [2][10]. Yet important challenges remain, and it is worth being precise about them: the weakness of prior work is less about missing capabilities than about how narrowly those capabilities have been grounded and evaluated.

It is tempting to frame the gap as the absence of a unified framework combining every component of Agentic AI into a single clinical decision support system, but that framing is too weak to defend, because integration on its own is not novel, MedAgents, Agent Hospital, and comparable systems already combine memory, reasoning, planning, and multi-agent collaboration [10][12]. The defensible gap lies in three specific places where even these integrated systems fall short: the data they are evaluated on, what their retrieval is grounded in, and whether verification and auditability are treated as first-class, measured components.

The first and most consequential shortcoming is evaluation on static examination question answering rather than longitudinal, real clinical records. MedAgents reports its gains on MedQA, MedMCQA, PubMedQA, and MMLU [10][31], and Agent Hospital, despite its evolving simulacrum of doctor and patient agents, ultimately benchmarks the acquired expertise on MedQA drawn from the USMLE [12][31]. AgentClinic moves toward interaction but with simulated rather than real patient trajectories [32]. Examination questions are curated and self-contained, whereas a real intensive-care admission is noisy, incompletely documented, and continually revised, so high exam accuracy offers little assurance that an agent can follow a deteriorating patient across days of MIMIC-IV observations [14]. Effective monitoring requires continuous access to previous admissions, laboratory results, medications, diagnoses, and treatment history, which examination benchmarks do not exercise.

The second shortcoming concerns what retrieval is grounded in. Several studies address hallucination through Retrieval-Augmented Generation, yet most implementations retrieve from guidelines, textbooks, or literature rather than from the

specific patient. MedRAG couples retrieval with a diagnostic knowledge graph but is oriented toward snapshot differential diagnosis and evaluated on datasets rather than a longitudinal ICU record [11]; Med-PaLM encodes clinical knowledge parametrically and does not retrieve from a patient timeline at all [2]; and EHRAgent is the clearest step toward EHR grounding, executing code over structured records, yet it frames the task as answering discrete questions rather than maintaining a running, evidence-linked model of a patient over an admission [33]. What is missing is retrieval anchored to the patient timeline, combining structured EHR evidence with medical literature so that patient-specific evidence enters the decision. The literature also exposes limits in reasoning reliability: Chain-of-Thought prompting, ReAct, and collaborative multi-agent reasoning improve decision quality [20][5][10], but hallucinated responses remain a considerable concern, and many systems still produce recommendations without verifiable evidence, which reduces clinician confidence. A related structural issue is that many healthcare AI systems operate as standalone models rather than collaborative multi-agent environments, so multi-agent collaboration for real-world clinical decision support remains an emerging rather than an established practice [6][18][10].

The third and most under-served shortcoming is trustworthiness, specifically verification and auditability as first-class, evaluated components. Recent work emphasizes explainability, fairness, and safety [2][30], but rarely includes a dedicated verification step that checks a generated recommendation against the retrieved patient evidence, and more rarely still measures whether the resulting audit trail is faithful to the evidence the system actually used. In a domain where a wrong recommendation can harm a patient, agent consensus is not equivalent to evidential entailment, and an audit trail that is merely produced is not the same as one whose faithfulness has been verified.

A comparative view makes this concrete. When systems are scored on general capabilities, memory, planning, reasoning, multi-agent collaboration, healthcare focus, and trustworthiness, the rows for ReAct, AutoGen, MetaGPT, MedAgents, Agent Hospital, and MedRAG collapse into nearly identical marks, which is precisely why an integration-based claim to novelty is weak. When the same systems are scored on five differentiating dimensions, patient-timeline RAG grounded in the EHR, a dedicated verification gate, longitudinal per-patient memory, evaluation on real ICU data such as MIMIC-IV, and a faithful audit trail, no prior system carries more than a single mark, and

most carry none [10][12][11][33][32][14]. Each prior system misses at least one of these dimensions, so the gap is real rather than rhetorical.

[Table 2.2: Research gap matrix (insert table here)]

These observations point to the need for an Agentic AI framework that grounds retrieval in the patient's own longitudinal EHR timeline, coordinates specialized clinical agents, and treats verification and faithful auditability as evaluated components rather than aspirations, supporting intelligent monitoring while providing transparent, evidence-based recommendations that assist clinicians without replacing their judgment. The proposed research addresses these gaps by designing such a framework using the MIMIC-IV dataset [14], combining longitudinal patient memory, ReAct-based reasoning [5], patient-timeline Retrieval-Augmented Generation over EHR data and medical literature, coordinated specialized agents, a dedicated verification gate, and human-in-the-loop validation to improve the quality, transparency, and reliability of AI-assisted clinical decision making. The architecture that realizes these components is presented in detail in Chapter 3.

2.11 Chapter Summary

This chapter reviewed the literature on Large Language Models, Agentic AI, Retrieval-Augmented Generation, multi-agent systems, and healthcare artificial intelligence, and read it critically to locate a defensible research gap. It traced the evolution of LLMs into LLM-based agents and then into Agentic AI, and examined the architectural components, memory, planning, reasoning, tool use, and multi-agent collaboration, that convert a passive model into a goal-directed system [3][4]. It analyzed influential agent frameworks, including ReAct, AutoGen, CAMEL, MetaGPT, Toolformer, Reflexion, Voyager, and Generative Agents, organizing their capabilities into a taxonomy and showing that each supplies a mechanism a clinical system needs while none supplies the whole [5][6][7][18][23][19][24][8]. Turning to the clinical domain, it reviewed artificial intelligence and Large Language Models in healthcare, clinical decision support, Retrieval-Augmented Generation, and trustworthy-AI mechanisms, examining systems such as Med-PaLM, Med-PaLM 2, Med-PaLM M, Clinical Camel, MedAgents, MedRAG, Agent Hospital, and EHRAgent [2][26][25][13][10][11][12][33].

The recurring finding is that existing research has advanced the individual capabilities, autonomous reasoning, knowledge retrieval, and collaborative agents, while leaving their integration under clinical supervision unrealized. Current approaches are evaluated largely on static examination benchmarks rather than longitudinal ICU records, ground retrieval in general knowledge rather than the specific patient's timeline, and rarely treat verification and faithful auditability as measured components. These gaps motivate the proposed Agentic AI Framework for Intelligent Patient Monitoring and Clinical Decision Support, which grounds retrieval in the patient's longitudinal EHR timeline, coordinates specialized agents with persistent memory and ReAct-based reasoning, and embeds a verification gate and human-in-the-loop validation, all grounded in the MIMIC-IV dataset [14][5][15]. The findings established here provide the theoretical and technical foundation for the research methodology and proposed architecture presented in the next chapter.

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